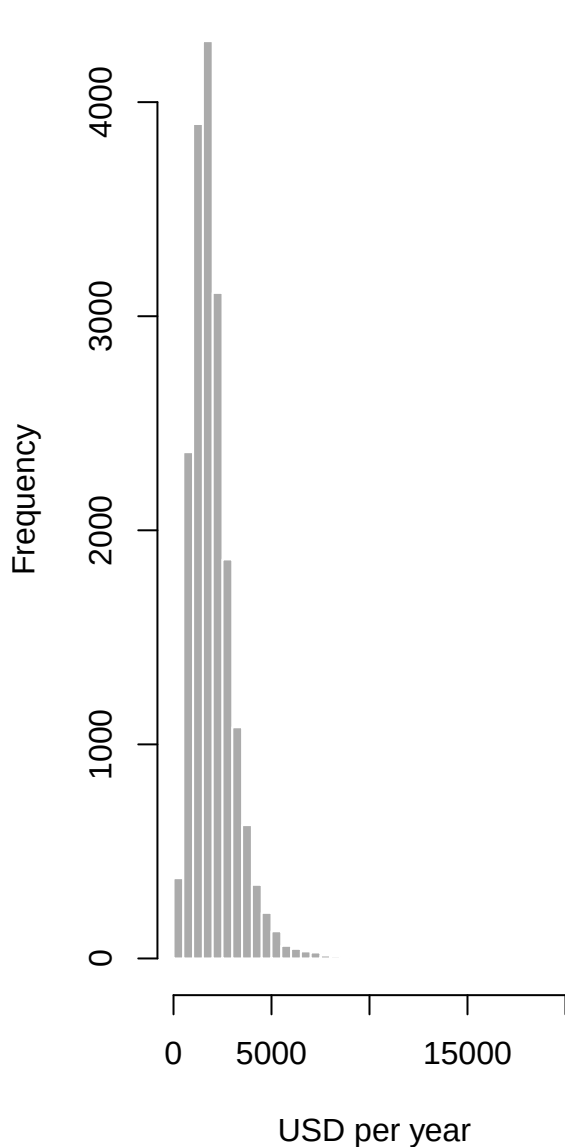
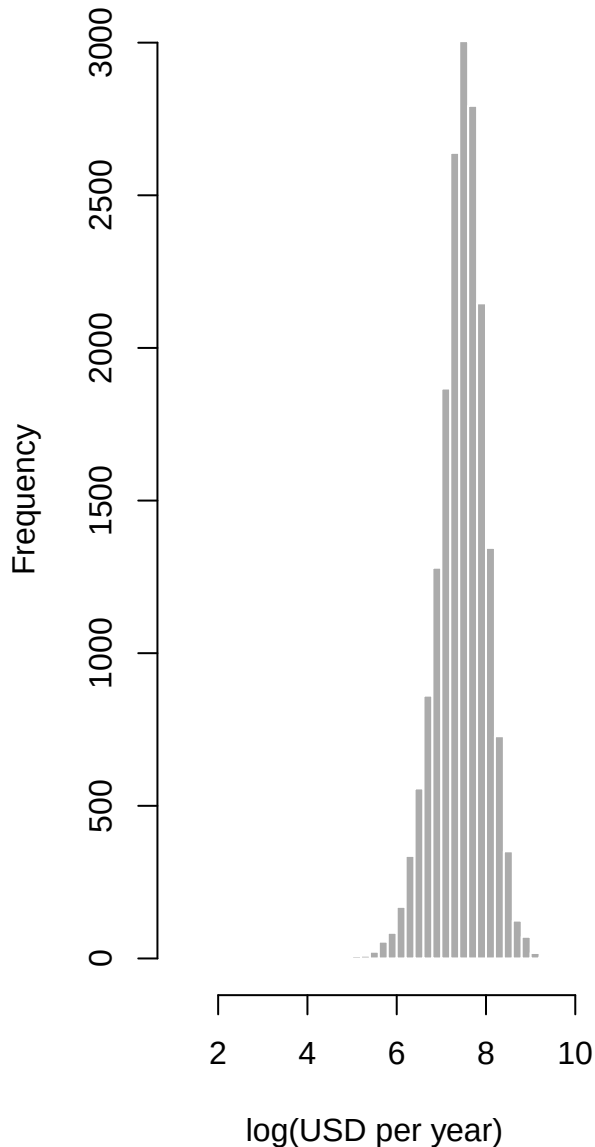


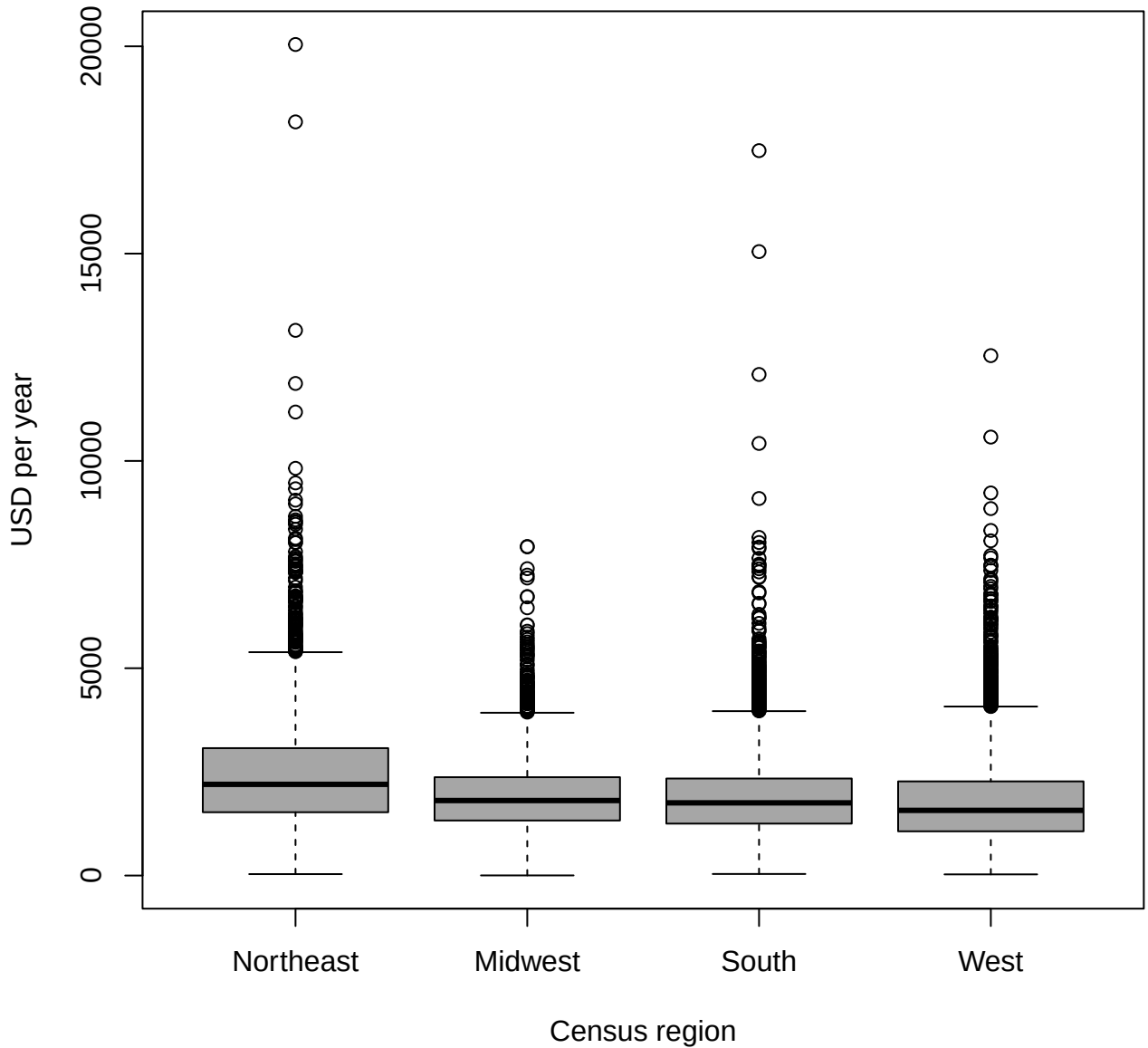
Annual energy cost



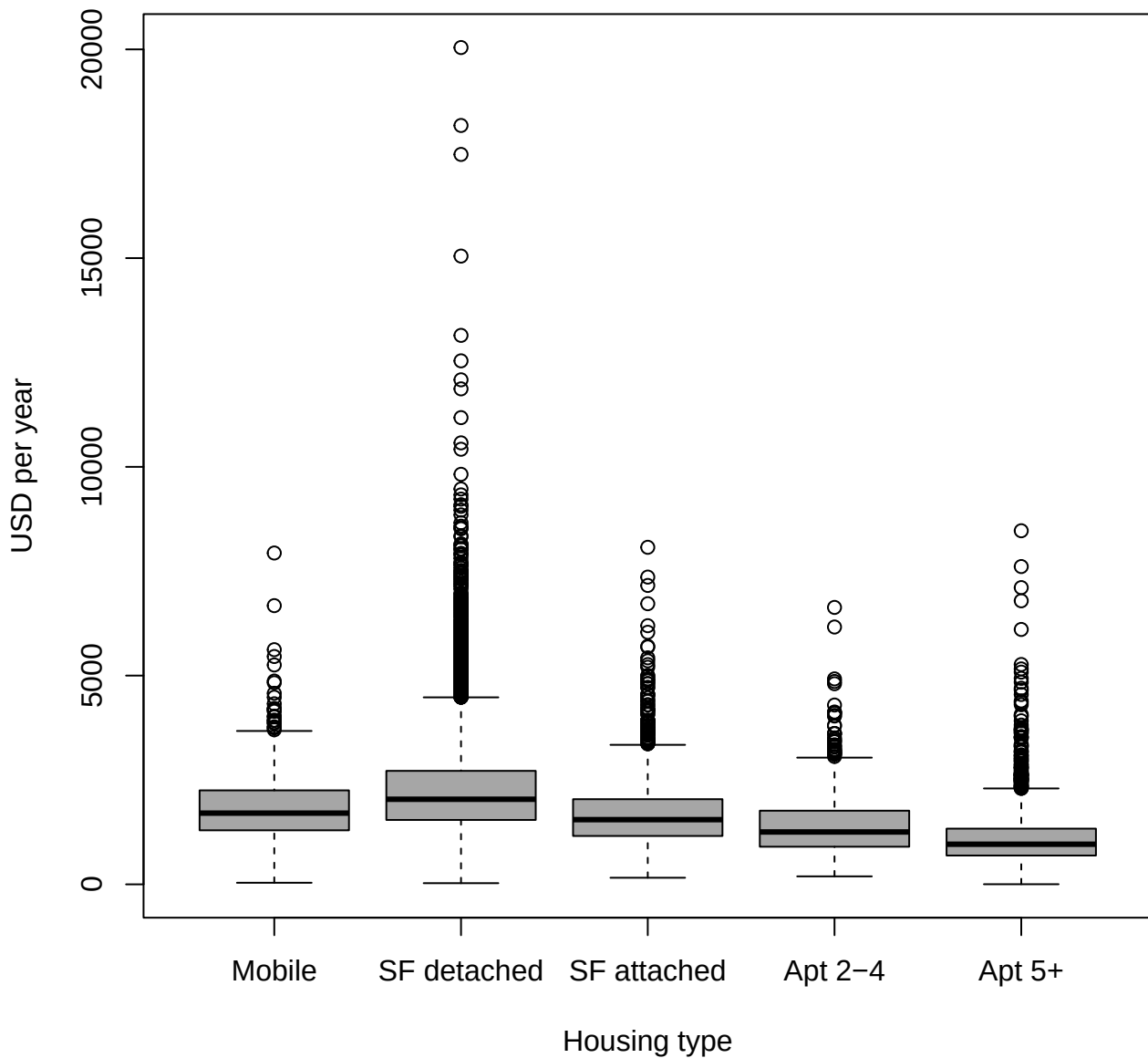
Annual energy cost, log scaled



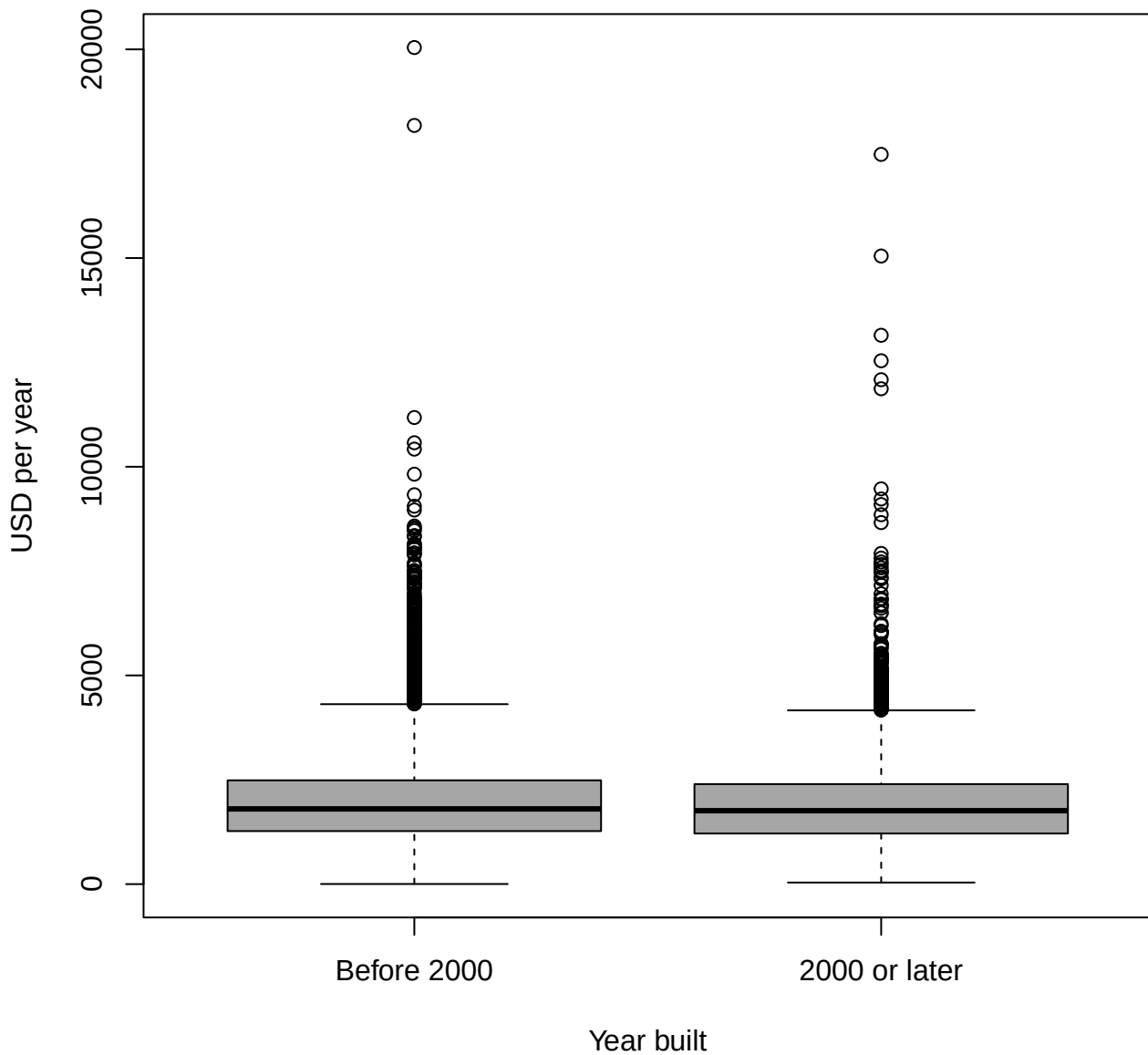
Annual energy cost by census region



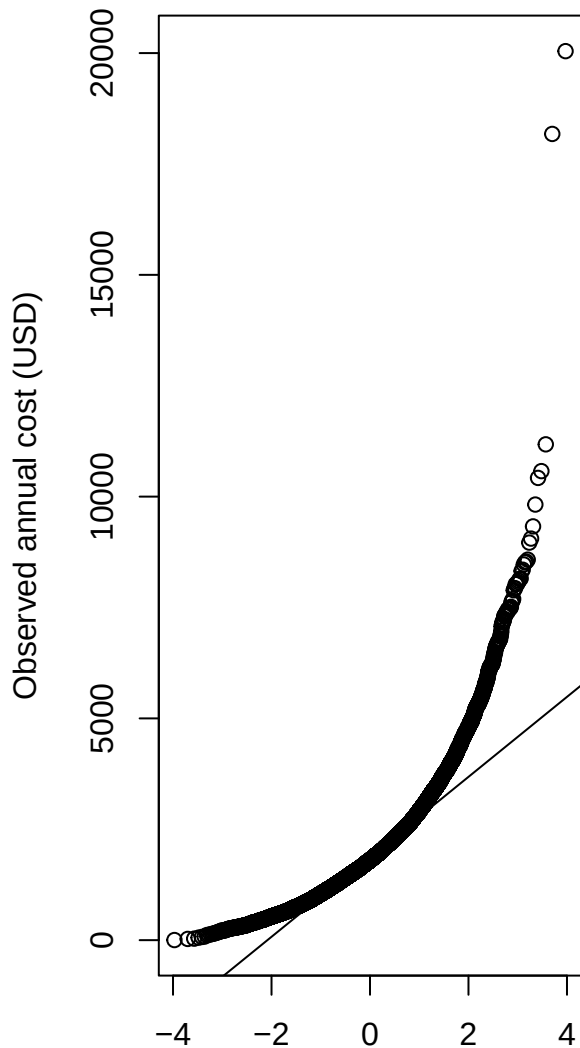
Annual energy cost by housing type



Annual energy cost by year built



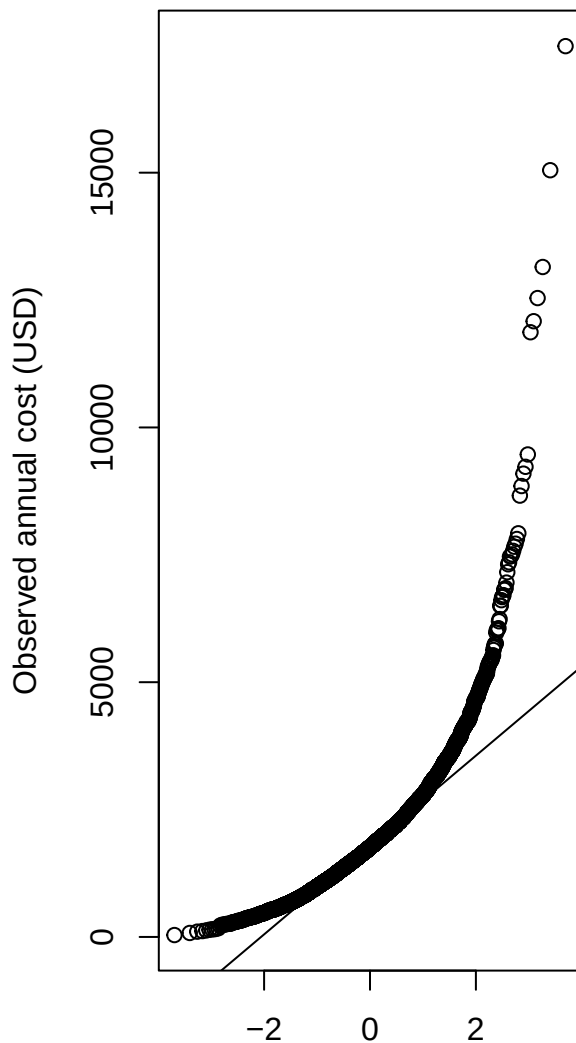
Homes built Before 2000



Theoretical quantiles (normal)

Points on the line = normally distributed

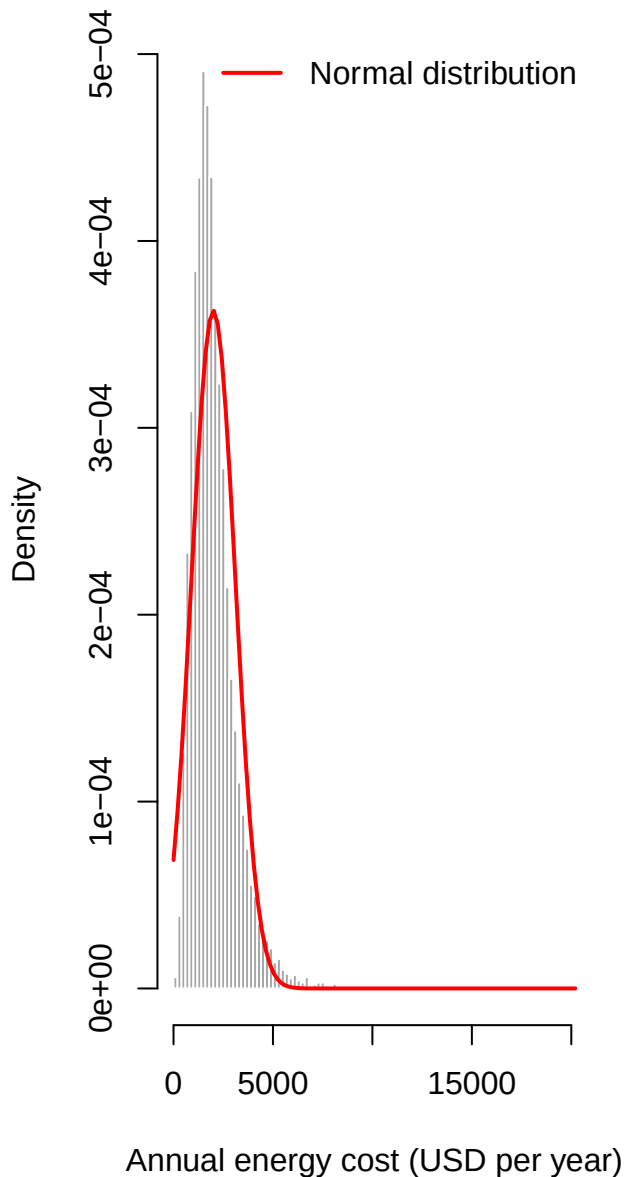
Homes built 2000 or later



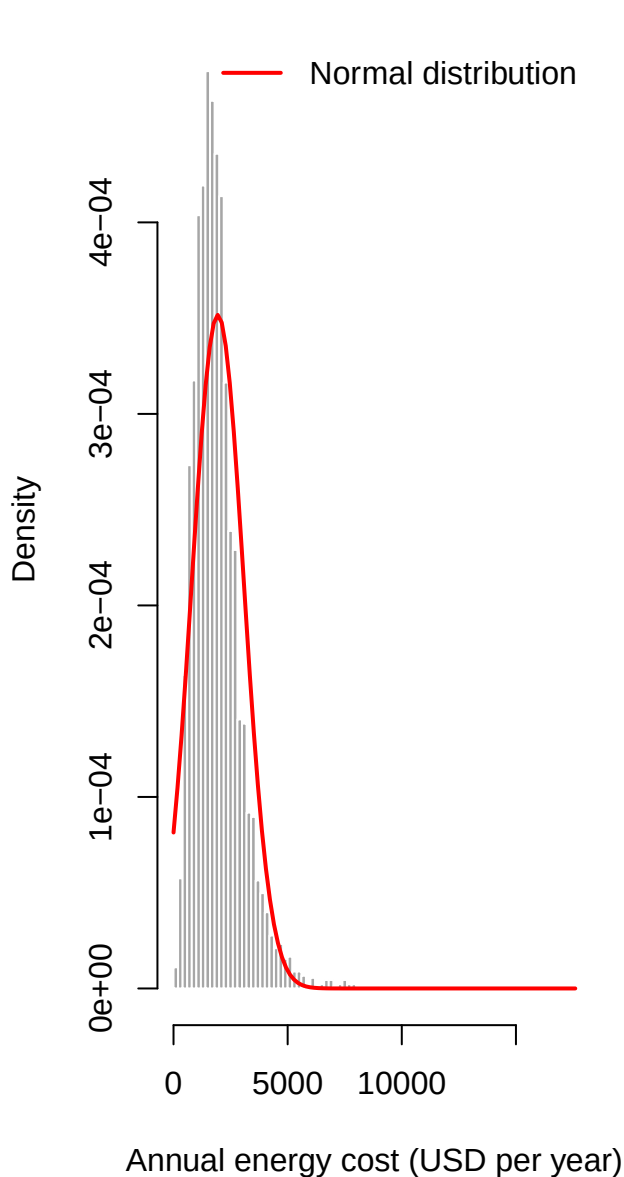
Theoretical quantiles (normal)

Points on the line = normally distributed

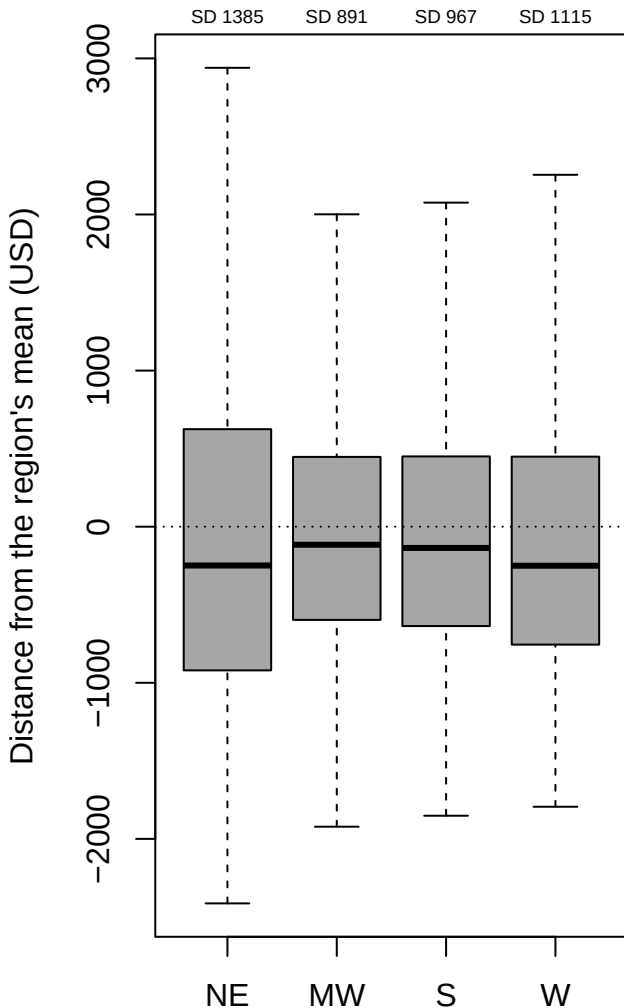
Homes built Before 2000



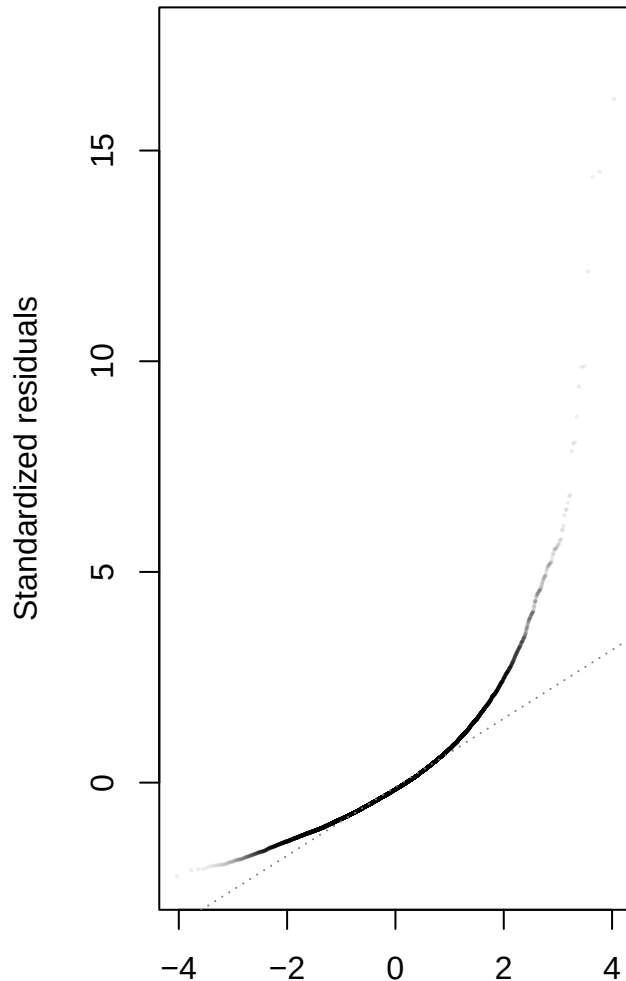
Homes built 2000 or later



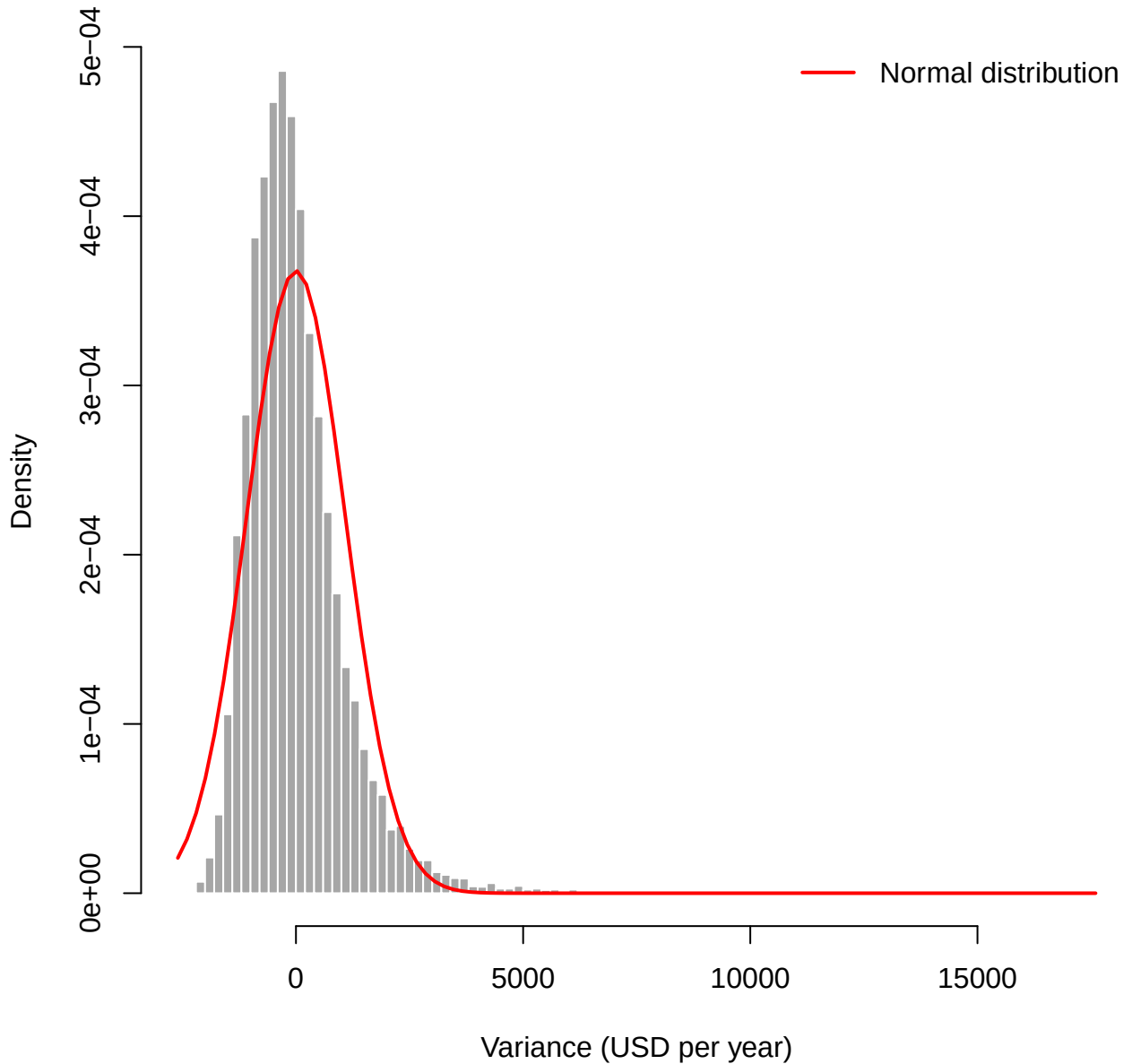
Spread even in each region?



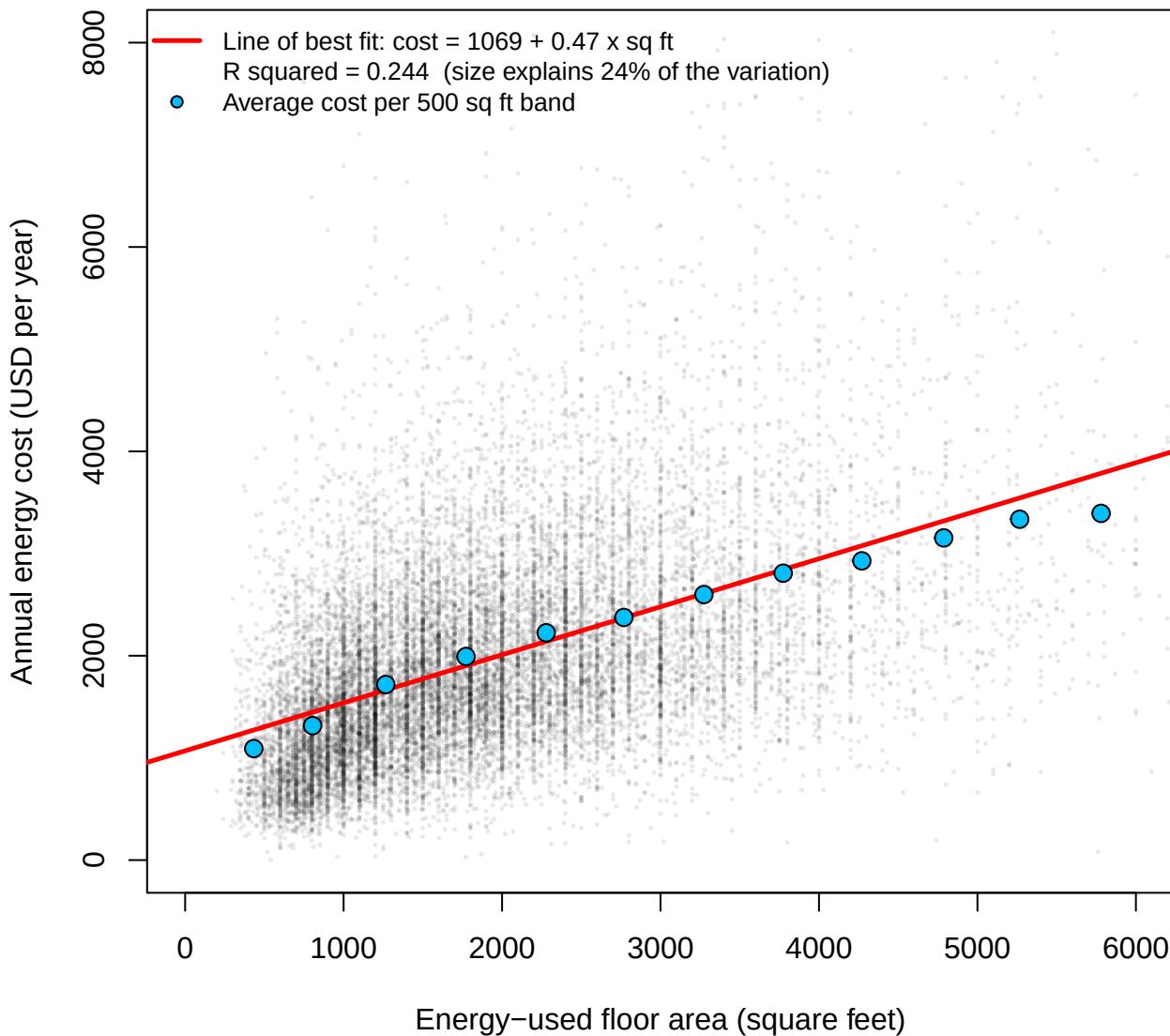
Are the errors normal?



ANOVA Variances against a normal distribution



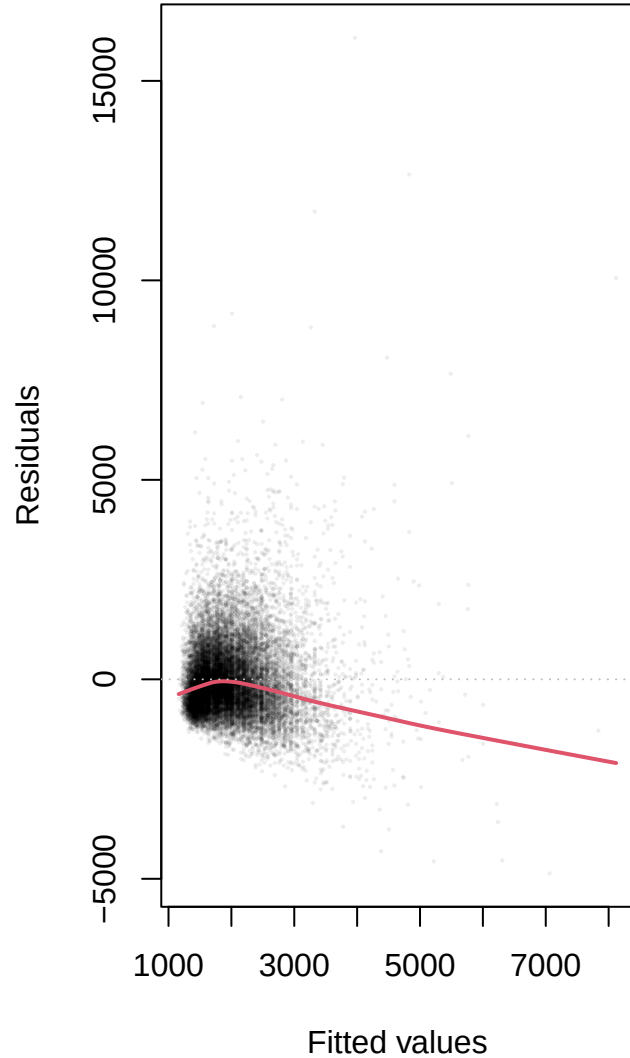
Larger homes cost more



Each dot is one household.

The vertical spread widens with size

Is the spread even?

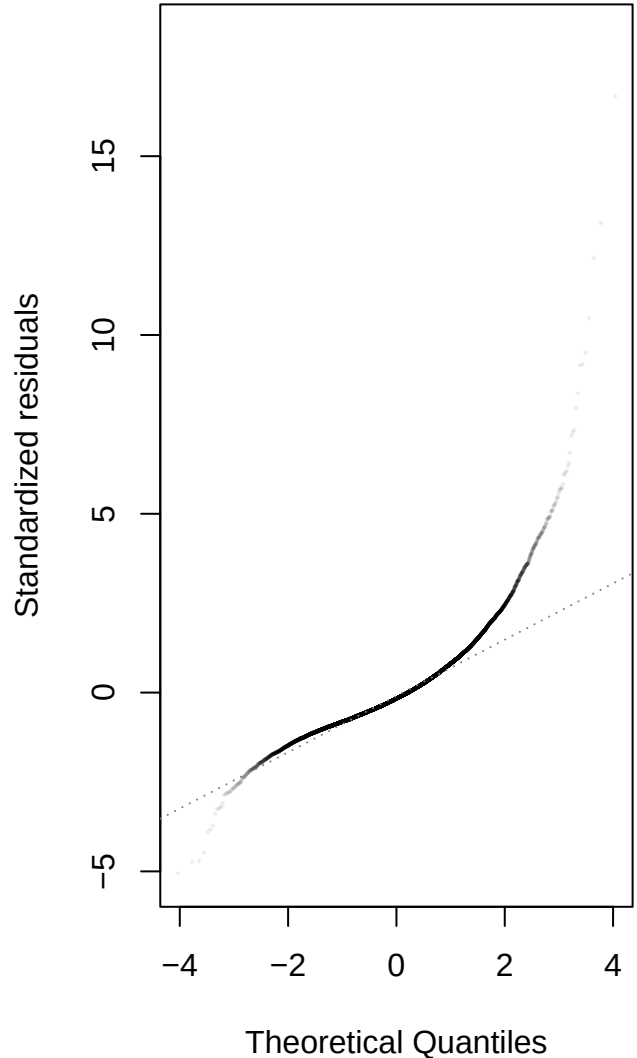


x = cost the model predicts for that home
y = how far the household sits from it

An even band = good

A widening cone = spread grows with size

Are the errors normal?

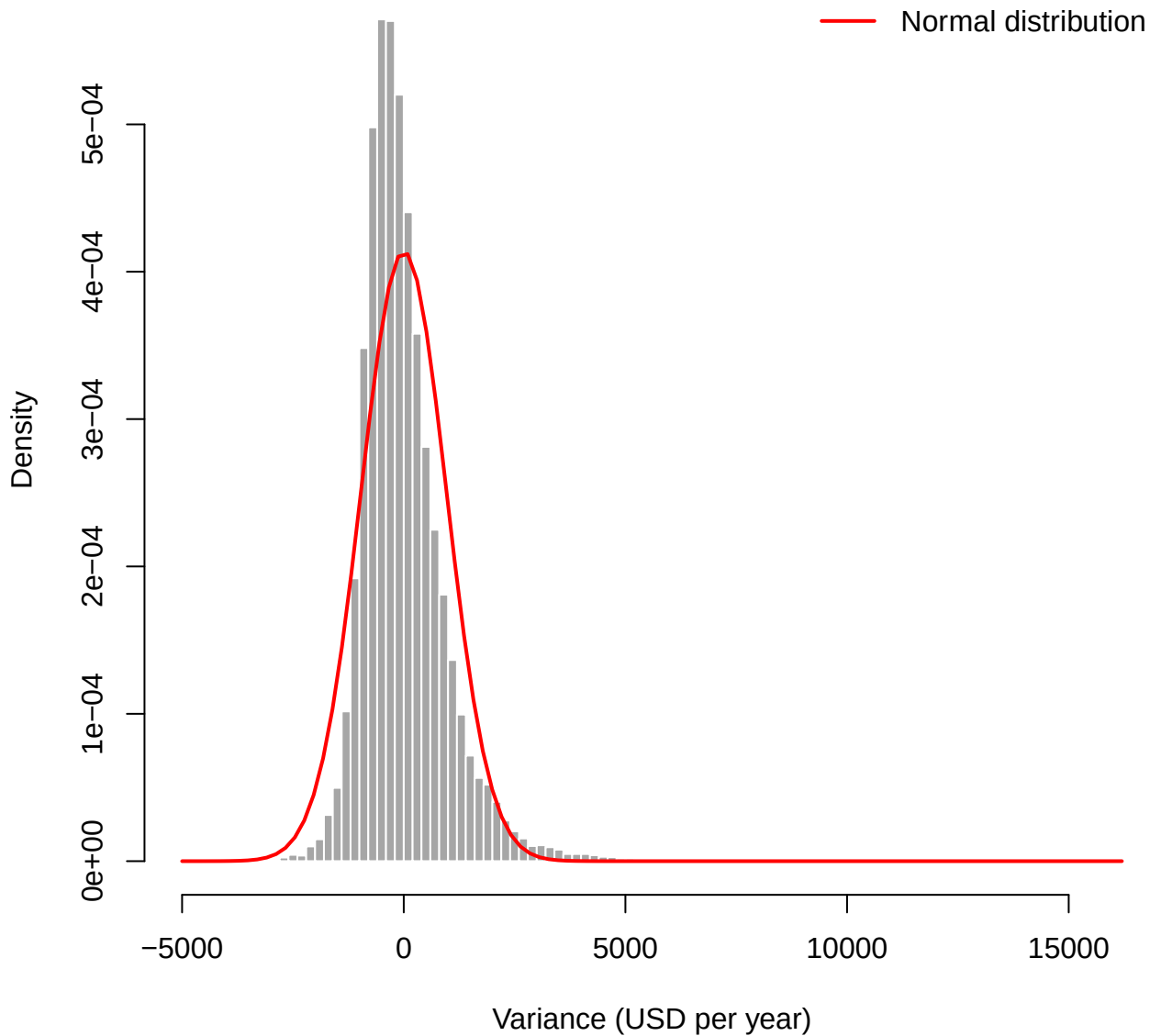


x = where a normal value would sit
y = where it actually sits

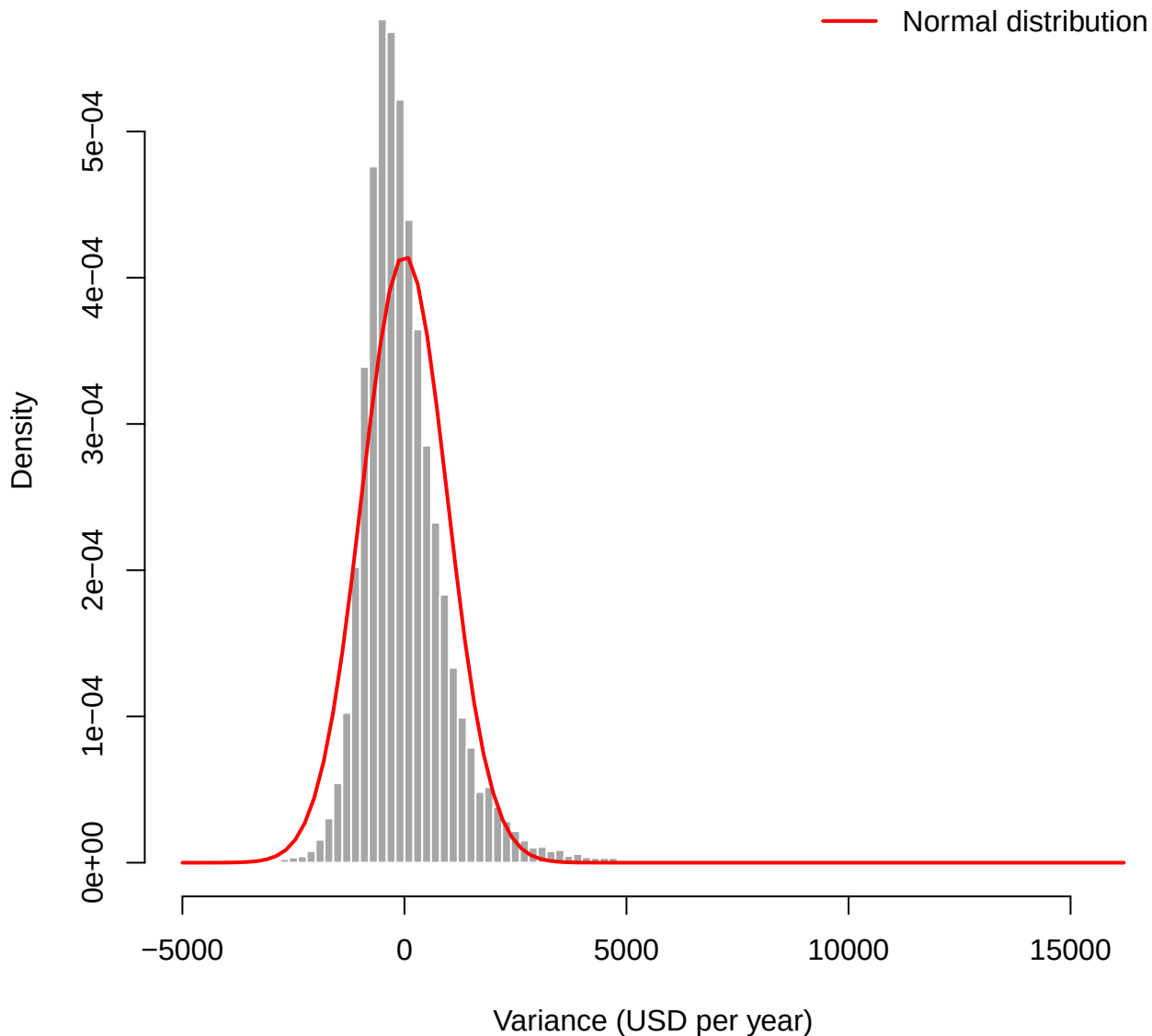
On the dashed line = normal

Curving up at the right = costs underpredicted

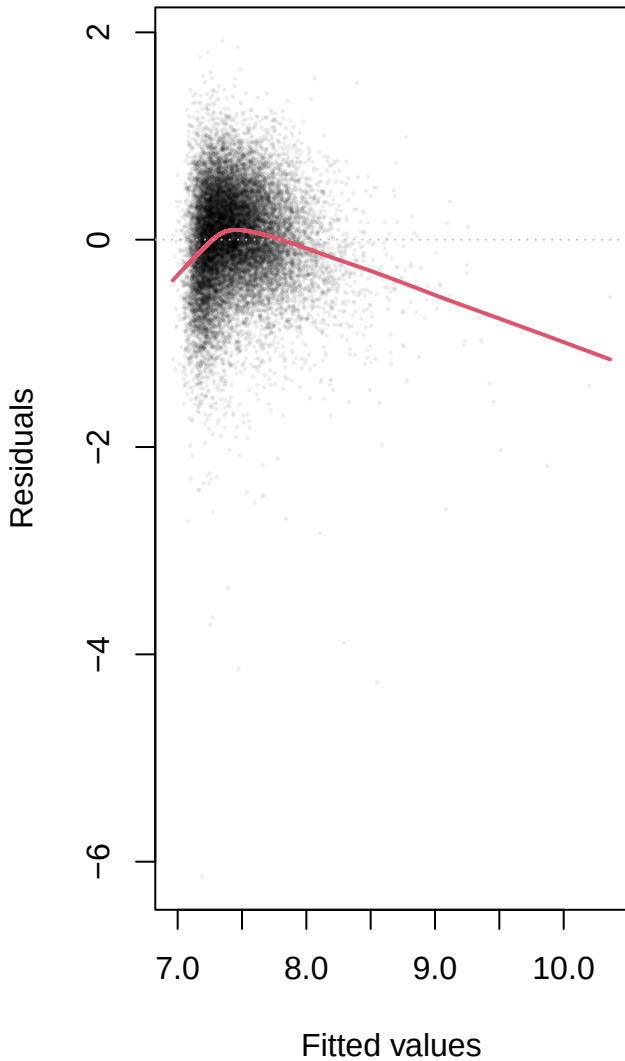
Regression Variances against a normal distribution



Raw model Variances against a normal curve

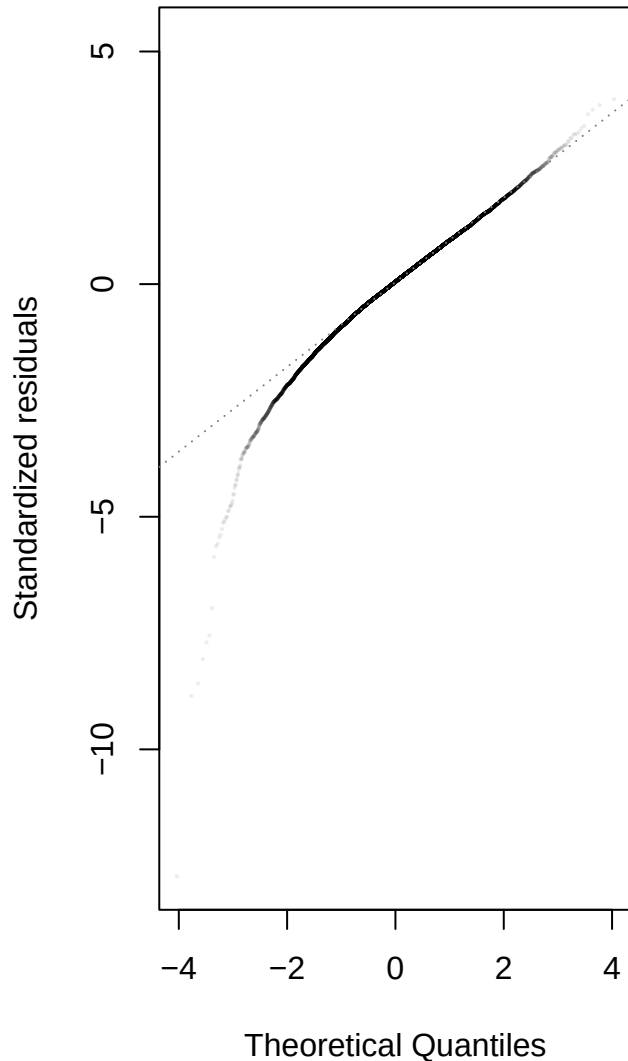


Log cost: spread even?



x = log cost the model predicts
y = how far the household sits from it

Log cost: errors normal?



x = where a normal value would sit
y = where it actually sits

Log model Variances against a normal curve

